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Multiple Input Multiple Output Wireless Communications: Introduction to MIMO Communications, MIMO system Model, MIMO Zero Forcing Receiver, MIMO MMSE Receiver, Singular Value decomposition of MIMO Channel, SVD and MIMO capacity, Alamouti and Space-Time Codes, Nonlinear MIMO receiver: V-Blast, MIMO Beamforming.

[Text1:6.1,6.2, 6.3, 6.4, 6.5, 6.6, 6.8, 6.9, 6.10]

Suggested Learning Resources:

Text Book

1. Aditya K Jagannatham, “Principles of Modern Wireless Communication systems, Theory and Practice”, Mc Graw Hill Education (India) Private Limited, 2017, ISBN 978-81- 265-4231-4.
2. Arunabha Ghosh, Jun Zhang, Jeffrey G. Andrews, Rias Muhamed, “Fundamentals of LTE”, Pearson India Education Services Private Limited, 2018, ISBN: 978-93-530-6239-2.

Reference Books

1. T L Singal, “Wireless Communications”, Mc Graw Hill Education (India) Private Limited, 2016, ISBN:978-0-07-068178-1
2. Theodore Rappaport, Wireless Communications: Principles and Practice, 2nd Edition, Prentice Hall Communications Engineering and Emerging Technologies Series, 2002, ISBN 0-13-042232-0.

Multiple-Input Multiple-Output Wireless Communications

6.1 Introduction to MIMO Wireless Communications

- Multiple-Input Multiple-Output (MIMO) wireless communications employ multiple antennas at the transmitter and the receiver.
- MIMO systems have multiple antennas, they can be employed to increase the reliability of the signal through diversity combining. This leads to diversity gain and a net decrease in the bit-error rate of the wireless communication system.
- MIMO wireless systems is that they enable a several-fold increase in the data rate of the wireless communication system by transmitting several information streams in parallel. This is termed

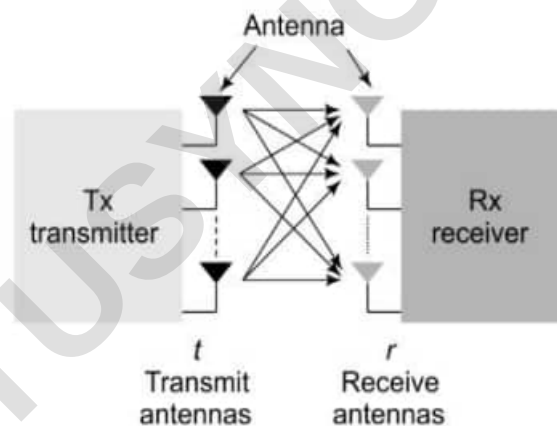


Figure 6.1 MIMO system schematic

spatial multiplexing.

6.2 MIMO System Model

- Consider a MIMO wireless system with t transmit antennas and r receive antennas. Such a MIMO system is also termed an $r \times t$ system.
- Let $x_1, x_2, \dots, x_t$ denote the t symbols transmitted from the t transmit antennas in the MIMO system, i.e., x_i denotes the symbol transmitted from the i^{th} transmit antenna $1 \leq i \leq t$. These transmit symbols can be stacked to form the t -dimensional vector, also termed the transmit vector,

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_t \end{bmatrix}$$

Corresponding to this transmission, let $y_1, y_2, \dots, y_r$ denote the r received symbols across the r receive antennas in the MIMO systems,

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_r \end{bmatrix}$$

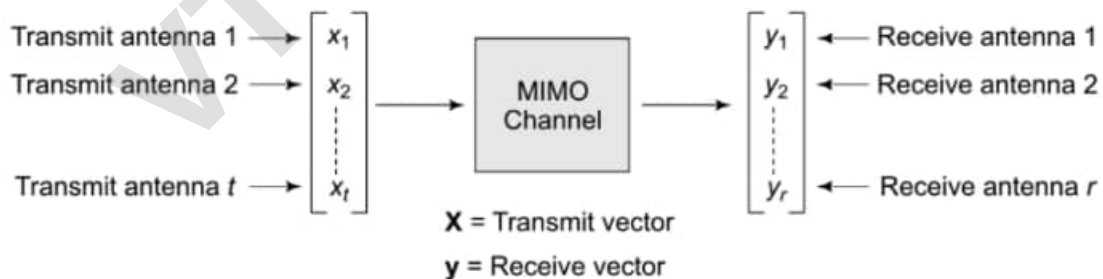


Figure 6.2 MIMO system input-output schematic

- Let the complex coefficient h^{ij} represent the fading channel coefficient between the i^{th} receive antenna and the j^{th} transmit antenna.
- Thus, there are a net of rt channel coefficients in this wireless scenario corresponding to all possible combinations of the r receive antennas and t transmit antennas. These can be arranged in a

matrix form as

$$\mathbf{H} = \begin{bmatrix} h_{11} & h_{12} & \dots & h_{1t} \\ h_{21} & h_{22} & \dots & h_{2t} \\ \vdots & \vdots & \ddots & \vdots \\ h_{r1} & h_{r2} & \dots & h_{rt} \end{bmatrix}$$

where the $r \times t$ dimensional matrix $\mathbf{H}$ is termed the MIMO channel matrix.

$$\underbrace{\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_r \end{bmatrix}}_{\mathbf{y}} = \underbrace{\begin{bmatrix} h_{11} & h_{12} & \dots & h_{1t} \\ h_{21} & h_{22} & \dots & h_{2t} \\ \vdots & \vdots & \ddots & \vdots \\ h_{r1} & h_{r2} & \dots & h_{rt} \end{bmatrix}}_{\mathbf{H}} \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_t \end{bmatrix}}_{\mathbf{x}} + \underbrace{\begin{bmatrix} n_1 \\ n_2 \\ \vdots \\ n_r \end{bmatrix}}_{\mathbf{n}}$$

- Let the additive noise at the receive antenna i be denoted by n_i , i.e., $n_1, n_2, \dots, n_r$ denote the
- additive noise at the r receive antennas. Thus, the net MIMO input output system model can be represented in vector form as
- This is succinctly represented using matrix notation as

$$\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{n}$$

the receive symbol y_1 is given as

$$y_1 = h_{11}x_1 + h_{12}x_2 + \dots + h_{1t}x_t + n_1$$

- from which it can be seen that all the symbols $x_1, x_2, \dots, x_t$ interfere at y_1 received at the

$$y_2 = h_{21}x_1 + h_{22}x_2 + \dots + h_{2t}x_t + n_2$$

receive antenna 1. Similarly, the receive symbol y_2 is given as

- This is, in general, true for all the receive antennas, i.e., at each receive antenna i , the receive symbol y_i is a linear of all the transmit symbols $x_1, x_2, \dots, x_t$ from the t transmit antennas, observed in additive noise n_i .
- Single-Input Multiple-Output (SIMO) For the special case of $t = 1$,

xsdcfvg

xcfvgbh

xcvb

$$\underbrace{\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_r \end{bmatrix}}_{\mathbf{y}} = \underbrace{\begin{bmatrix} h_1 \\ h_2 \\ \vdots \\ h_r \end{bmatrix}}_{\mathbf{h}} x + \underbrace{\begin{bmatrix} n_1 \\ n_2 \\ \vdots \\ n_r \end{bmatrix}}_{\mathbf{n}}$$

- Similarly, for the case of one receive antenna, i.e., $r = 1$ and multiple transmit antennas, it is termed a Multiple-Input Single-Output (MISO) system model

$$y = \underbrace{\begin{bmatrix} h_1 & h_1 & \dots & h_t \end{bmatrix}}_{\mathbf{h}^T} \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_t \end{bmatrix}}_{\mathbf{x}} + n$$

- Finally, for $r = t = 1$, i.e., a single receive and transmit antenna, it reduces to the single-input single-output (SISO) system, modelled as

$$\mathbf{y} = \mathbf{h}\mathbf{x} + \mathbf{n}$$

- The covariance matrix of the noise $\mathbf{R}_n$ of the noise vector $\mathbf{n}$ defined as

$$\mathbf{R}_n = \mathbf{E}\{\mathbf{n}\mathbf{n}^H\}$$

$$= \mathbf{E} \left\{ \begin{bmatrix} n_1 \\ n_2 \\ \vdots \\ n_L \end{bmatrix} \begin{bmatrix} n_1^* & n_2^* & \dots & n_L^* \end{bmatrix} \right\}$$

$$\begin{aligned}
&= \begin{bmatrix} E\{|n_1|^2\} & E\{n_1 n_2^*\} & \dots & E\{n_1 n_r^*\} \\ E\{n_2 n_1^*\} & E\{|n_2|^2\} & \dots & E\{n_2 n_r^*\} \\ \vdots & \vdots & \ddots & \vdots \\ E\{n_r n_1^*\} & E\{n_r n_2^*\} & \dots & E\{|n_r|^2\} \end{bmatrix} \\
&= \begin{bmatrix} \sigma_n^2 & 0 & 0 & \dots & 0 \\ 0 & \sigma_n^2 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & \sigma_n^2 \end{bmatrix} \\
&= \sigma_n^2 \mathbf{I}_r
\end{aligned}$$

- The noise vector $\mathbf{n}$ with the covariance structure above is termed spatially uncorrelated additive noise, since the noise samples at the different antennas i, j are independent, i.e

$$E\{n_i n_j^*\} = 0$$

- To denote the transmission and reception across different time instants, one can add the time index k to the MIMO system model to frame the net model as

$$\underbrace{\begin{bmatrix} y_1(k) \\ y_2(k) \\ \vdots \\ y_r(k) \end{bmatrix}}_{\mathbf{y}(k)} = \underbrace{\begin{bmatrix} h_{11} & h_{12} & \dots & h_{1t} \\ h_{21} & h_{22} & \dots & h_{2t} \\ \vdots & \vdots & \ddots & \vdots \\ h_{r1} & h_{r2} & \dots & h_{rt} \end{bmatrix}}_{\mathbf{H}} \underbrace{\begin{bmatrix} x_1(k) \\ x_2(k) \\ \vdots \\ x_t(k) \end{bmatrix}}_{\mathbf{x}(k)} + \underbrace{\begin{bmatrix} n_1(k) \\ n_2(k) \\ \vdots \\ n_r(k) \end{bmatrix}}_{\mathbf{n}(k)}$$

- Thus, the vectors $\mathbf{y}(k)$, $\mathbf{x}(k)$, $\mathbf{n}(k)$ define the receive, transmit, and noise vectors of the MIMO wireless communication system at the time instant k .

we have assumed the channel matrix $\mathbf{H}$ to be constant or, in other words, not dependent on the time instant k . This is also termed a slow fading or quasi-static channel matrix, indicating that the channel coefficients are constant over the block of MIMO vectors that are transmitted.

- assume that any two noise samples across two different time instants are uncorrelated,

i.e., $E \{ n_i(k) n_j^*(l) \} = 0$ if $k \neq l$. Hence, the noise covariance matrix is given as

$$E \{ \mathbf{n}(k) \mathbf{n}(l)^H \} = \sigma^2 \delta(k-l) \mathbf{I}_r$$



where the delta function $\delta(k-l) = 1$ if $k=l$ and 0 otherwise.

- This noise process, which is uncorrelated across different antennas and time instants is termed *spatio-temporally* uncorrelated noise.

6.3 MIMO Zero-Forcing (ZF) Receiver

- Now describe the process to recover the transmitted signal vector $\mathbf{x}$ from the received vector $\mathbf{y}$ at the MIMO receiver. This can be considered as solving the system of linear equations,

$$\mathbf{y} = \mathbf{H}\mathbf{x}$$

- Consider, where $r = t$, i.e., the number of receive antennas is equal to the number of transmit antennas. $\hat{\mathbf{x}}$ of the transmit vector $\mathbf{x}$ is given as

$$\hat{\mathbf{x}} = \mathbf{H}^{-1}\mathbf{y}$$

- However, frequently, one has more receive antennas than transmit antennas, i.e., $r > t$. In this scenario, the system $\mathbf{y} = \mathbf{H}\mathbf{x}$ is given as

$$\underbrace{\begin{bmatrix} y_1(k) \\ y_2(k) \\ \vdots \\ y_r(k) \end{bmatrix}}_{\mathbf{y}(k)} = \underbrace{\begin{bmatrix} h_{11} & h_{12} & \dots & h_{1t} \\ h_{21} & h_{22} & \dots & h_{2t} \\ \vdots & \vdots & \ddots & \vdots \\ \vdots & \vdots & \ddots & \vdots \\ h_{r1} & h_{r2} & \dots & h_{rt} \end{bmatrix}}_{\mathbf{H}} \underbrace{\begin{bmatrix} x_1(k) \\ x_2(k) \\ \vdots \\ x_t(k) \end{bmatrix}}_{\mathbf{x}(k)}$$

- The vector $\mathbf{x}$ which minimizes the estimation error $f(\hat{\mathbf{x}})$,

$$f(\mathbf{x}) = \|\mathbf{y} - \mathbf{H}\mathbf{x}\|^2$$

- The above error function is also termed the least-squares error function and the resulting estimator is termed the least-squares estimator.

- The above error function can be expanded as

$$\begin{aligned} f(\mathbf{x}) &= \|\mathbf{y} - \mathbf{H}\mathbf{x}\|^2, \\ &= (\mathbf{y} - \mathbf{H}\mathbf{x})^T (\mathbf{y} - \mathbf{H}\mathbf{x}) \\ &= (\mathbf{y}^T - \mathbf{x}^T \mathbf{H}^T) (\mathbf{y} - \mathbf{H}\mathbf{x}) \\ &= \mathbf{y}^T \mathbf{y} - \mathbf{x}^T \mathbf{H}^T \mathbf{y} - \mathbf{y}^T \mathbf{H}\mathbf{x} + \mathbf{x}^T \mathbf{H}^T \mathbf{H}\mathbf{x} \\ &= \mathbf{y}^T \mathbf{y} - 2\mathbf{x}^T \mathbf{H}^T \mathbf{y} + \mathbf{x}^T \mathbf{H}^T \mathbf{H}\mathbf{x} \end{aligned} \quad (6.1)$$

- we have used the relation

$$\mathbf{y}^T \mathbf{H}\mathbf{x} = (\mathbf{y}^T \mathbf{H}\mathbf{x})^T = \hat{\mathbf{x}}^T \mathbf{H}^T \mathbf{y}$$

This is due to the fact that $\mathbf{y}^T \mathbf{H}\mathbf{x}$ is a scalar and, hence, is equal to its transpose.

- To find the minimum of the error function $f(\mathbf{x})$ with respect to $\mathbf{x}$, we have to set the derivative with respect to $\mathbf{x}$ equal to 0.

Consider a multidimensional function $g(\mathbf{x})$. The vector derivative of $g(\mathbf{x})$ with respect to $\mathbf{x}$ is defined as

$$\frac{\partial g(\mathbf{x})}{\partial \mathbf{x}} = \begin{bmatrix} \frac{\partial g(\mathbf{x})}{\partial x_1} \\ \frac{\partial g(\mathbf{x})}{\partial x_2} \\ \vdots \\ \frac{\partial g(\mathbf{x})}{\partial x_t} \end{bmatrix}$$

- which is basically a t -dimensional vector with the i^{th} component equal to the derivative of $f(\mathbf{x})$ with respect to x_i . Consider any vector $\mathbf{c} = [c_1, c_2, \dots, c_t]^T$. Let the function $g(\mathbf{x})$ be defined as

$$g(\mathbf{x}) = \mathbf{c}^T \mathbf{x} = c_1 x_1 + c_2 x_2 + \dots + c_t x_t$$

Hence, it can be seen that, $\frac{\partial g(\mathbf{x})}{\partial x_i} = c_i$.

$$\frac{\partial g(\mathbf{x})}{\partial \mathbf{x}} = \begin{bmatrix} \frac{\partial g(\mathbf{x})}{\partial x_1} \\ \frac{\partial g(\mathbf{x})}{\partial x_2} \\ \vdots \\ \frac{\partial g(\mathbf{x})}{\partial x_t} \end{bmatrix} = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_t \end{bmatrix} = \mathbf{c}$$

In fact, it can also be seen that since $\mathbf{c}^T \mathbf{x} = \mathbf{x}^T \mathbf{c}$, we have

$$\frac{\partial \mathbf{c}^T \mathbf{x}}{\partial \mathbf{x}} = \frac{\partial \mathbf{x}^T \mathbf{c}}{\partial \mathbf{x}} = \mathbf{c}$$

Going back to the expansion of the error function $f(\mathbf{x})$ from Eq. (6.1), it can be seen that the derivative of each component with respect to $\mathbf{x}$ can be computed as follows. Consider the quantity $\mathbf{y}^T \mathbf{y} = \|\mathbf{y}\|^2$. Observe that this does not depend on $\mathbf{x}$. Hence, we have $\frac{\partial \mathbf{y}^T \mathbf{y}}{\partial \mathbf{x}} = 0$. Consider the component $2\mathbf{x}^T \mathbf{H}^T \mathbf{y}$. This is in the form of $\mathbf{x}^T \mathbf{c}$, where $\mathbf{c} = 2\mathbf{H}^T \mathbf{y}$. Hence, the derivative of this component with respect to $\mathbf{x}$ is given as $\frac{\partial (2\mathbf{x}^T \mathbf{H}^T \mathbf{y})}{\partial \mathbf{x}} = 2\mathbf{H}^T \mathbf{y}$. Now, consider the last component $\mathbf{x}^T \mathbf{H}^T \mathbf{H} \mathbf{x}$. This can be differentiated employing the product rule as

$$\begin{aligned} \frac{\partial (\mathbf{x}^T \mathbf{H}^T \mathbf{H} \mathbf{x})}{\partial \mathbf{x}} &= \mathbf{H}^T \mathbf{H} \mathbf{x} + (\mathbf{x}^T \mathbf{H}^T \mathbf{H})^T \\ &= 2\mathbf{H}^T \mathbf{H} \mathbf{x} \end{aligned}$$

- the derivative for the error function $f(\mathbf{x})$ can be simplified as

$$\frac{\partial f(\mathbf{x})}{\partial \mathbf{x}} = -2\mathbf{H}^T \mathbf{y} + 2\mathbf{H}^T \mathbf{H} \mathbf{x}$$

- At the optimal estimate of the transmit vector $\hat{\mathbf{x}}$ where the above error is minimized, we must

have the derivative equal to 0. Using this condition,

$$\begin{aligned}\left. \frac{\partial f(\mathbf{x})}{\partial \mathbf{x}} \right|_{\mathbf{x}=\hat{\mathbf{x}}} &= 0 \\ -2\mathbf{H}^T \mathbf{y} + 2\mathbf{H}^T \mathbf{H} \hat{\mathbf{x}} &= 0 \\ \mathbf{H}^T \mathbf{H} \hat{\mathbf{x}} &= \mathbf{H}^T \mathbf{y} \\ \hat{\mathbf{x}} &= (\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T \mathbf{y}\end{aligned}$$

- Finally, for the case of complex vectors/matrices $\mathbf{x}$, $\mathbf{y}$, $\mathbf{H}$, the transpose in the above expression can be replaced by the Hermitian operator to yield

$$\hat{\mathbf{x}} = (\mathbf{H}^H \mathbf{H})^{-1} \mathbf{H}^H \mathbf{y}$$

- The zero-forcing decoder can be expressed as

$$\hat{\mathbf{x}}_{ZF} = \underbrace{(\mathbf{H}^H \mathbf{H})^{-1} \mathbf{H}^H}_{\mathbf{F}_{ZF}} \mathbf{y} \quad (6.2)$$

The matrix $\mathbf{F}_{ZF} = (\mathbf{H}^H \mathbf{H})^{-1} \mathbf{H}^H$ is also termed the *zero-forcing receiver matrix* and the estimate $\hat{\mathbf{x}}_{ZF}$ is, therefore, given as

$$\hat{\mathbf{x}}_{ZF} = \mathbf{F}_{ZF} \mathbf{y}$$

6.3.1 Properties of the Zero-Forcing Receiver Matrix $\mathbf{F}_{ZF}$

Consider the matrix product $\mathbf{F}_{ZF} \mathbf{H}$, which can be simplified as

$$\begin{aligned}\mathbf{F}_{ZF} \mathbf{H} &= (\mathbf{H}^H \mathbf{H})^{-1} \mathbf{H}^H \mathbf{H} \\ &= (\mathbf{H}^H \mathbf{H})^{-1} (\mathbf{H}^H \mathbf{H}) = \mathbf{I}_t\end{aligned} \quad (6.3)$$

- where $\mathbf{I}_t$ denotes the identity matrix of dimension t . Thus, multiplying the matrix $\mathbf{F}_{ZF}$ with the channel matrix indeed produces the identity matrix.
- For instance, if $\mathbf{F}_{ZF}$ to be the inverse of $\mathbf{H}$, it must also satisfy the property that $\mathbf{H} \mathbf{F}_{ZF} = \mathbf{I}$. However, we have,

$$\mathbf{H} \mathbf{F}_{ZF} = \mathbf{H} (\mathbf{H}^H \mathbf{H})^{-1} \mathbf{H}^H \neq \mathbf{I}_t$$

- if $r = t$ and the matrix $\mathbf{H}$ is invertible then the pseudo-inverse is actually equal to the inverse. This can be seen as follows.
- Thus, for this particular case, the pseudo-inverse reduces to the matrix inverse and is unique.

$$\begin{aligned}\mathbf{F}_{ZF} &= \mathbf{H} (\mathbf{H}^H \mathbf{H})^{-1} \mathbf{H}^H \\ &= \mathbf{H}^{-1} (\mathbf{H}^H)^{-1} \mathbf{H}^H \\ &= \mathbf{H}^{-1}\end{aligned}$$

6.3.2 Principle of Orthogonality Interpretation of ZF Receiver

- we present an intuitive-reasoning-based approach to derive the zero-forcing decoder above.

Denoting the column $\mathbf{h}_i = [h_{1i}, h_{2i}, \dots, h_{ri}]^T$ we denote the i^{th} column of the channel matrix $\mathbf{H}$. Then, the system of equations above can be succinctly represented as

$$\underbrace{\begin{bmatrix} \vdots \\ y_r(k) \\ \vdots \end{bmatrix}}_{\mathbf{y}(k)} = \underbrace{\begin{bmatrix} \vdots & \vdots & \ddots & \vdots \\ \vdots & \vdots & \ddots & \vdots \\ h_{r1} & h_{r2} & \dots & h_{rt} \end{bmatrix}}_{\mathbf{H}} \underbrace{\begin{bmatrix} \vdots \\ x_t(k) \end{bmatrix}}_{\mathbf{x}(k)}$$

$\hat{\mathbf{y}} = x_1 \mathbf{h}_1 + x_2 \mathbf{h}_2 + \dots + x_t \mathbf{h}_t$

Consider once again the system of equations at the receiver for $r > t$.

- where $\hat{\mathbf{y}}$ is the approximation of $\mathbf{y}$ and we have to minimize $\|\mathbf{y} - \hat{\mathbf{y}}\|^2$. Observe that there are t columns of the channel matrix $\mathbf{H}$, which are $\mathbf{h}_1, \mathbf{h}_2, \dots, \mathbf{h}_t$.
- Hence, they represent a t -dimensional subspace. However, the vector $\mathbf{y}$ can lie anywhere in the r -dimensional space, where approximation error $\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}}$ the error is shown for different choices of $\hat{\mathbf{y}}$.
- Observe that the error vector $\mathbf{e}$ is orthogonal to $\mathbf{h}_i$ if $\mathbf{h}_i^H \mathbf{e} = 0$. Therefore, since $\mathbf{e}$ is orthogonal to

the subspace spanned by the columns of $\mathbf{H}$, it follows that it is orthogonal to each of the columns

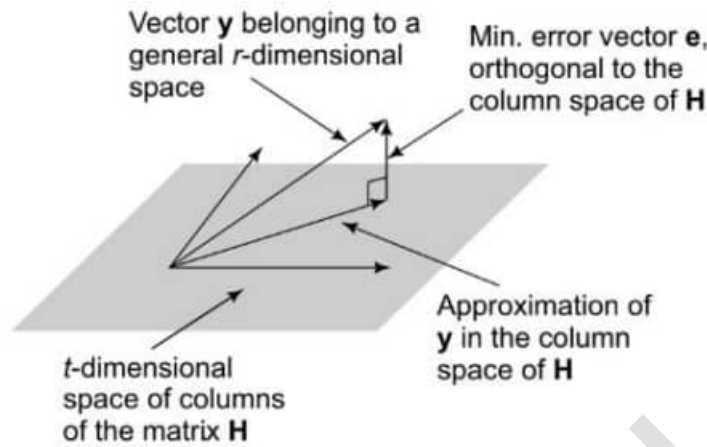


Figure 6.3 Zero-forcing: principle of orthogonality

of $\mathbf{H}$

➤ Hence, we have

$$\mathbf{h}_1^H \mathbf{e} = 0$$

$$\mathbf{h}_2^H \mathbf{e} = 0$$

$$\vdots$$

$$\mathbf{h}_t^H \mathbf{e} = 0,$$

from which it follows

$$\mathbf{H}^H \mathbf{e} = \mathbf{0}_{t \times 1}$$

➤ We now employ the above principle of orthogonality to derive the expression for the zero-forcing estimate $\hat{\mathbf{x}}_{\text{ZF}}$ of the transmitted symbol vector $\mathbf{x}$. This can be derived by substituting the expression $\mathbf{e} = \mathbf{y} - \mathbf{H}\hat{\mathbf{x}}_{\text{ZF}}$ in the above result as

$$\mathbf{H}^H \underbrace{(\mathbf{y} - \mathbf{H}\hat{\mathbf{x}}_{\text{ZF}})}_{\mathbf{e}} = 0$$

$$\mathbf{H}^H \mathbf{y} - \mathbf{H}^H \mathbf{H} \hat{\mathbf{x}}_{\text{ZF}} = 0$$

$$\mathbf{H}^H \mathbf{y} = \mathbf{H}^H \mathbf{H} \hat{\mathbf{x}}_{\text{ZF}}$$

$$\hat{\mathbf{x}}_{\text{ZF}} = (\mathbf{H}^H \mathbf{H})^{-1} \mathbf{H}^H \mathbf{y}$$

➤ which is exactly identical to the expression for the zero-forcing MIMO decoder derived earlier

in Eq. (6.2). Thus, the principle of orthogonality can be conveniently employed to deduce the expression for the optimal ZF MIMO decoder while also giving valuable insights into the

EXAMPLE 6.1

Compute the MIMO zero-forcing receiver for the channel matrix $\mathbf{H}$ given as

$$\mathbf{H} = \begin{bmatrix} 2 & 3 \\ 1 & 3 \\ 4 & 2 \end{bmatrix} \quad (6.4)$$

decoding procedure.

Solution: It can be seen that the MIMO channel matrix above is of size 3×2 , implying that the number of receive antennas is $r = 3$, while the number of transmit antennas is $t = 2$. Thus, the MIMO system model can be described as

$$\underbrace{\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}}_{\mathbf{y}} = \underbrace{\begin{bmatrix} 2 & 3 \\ 1 & 3 \\ 4 & 2 \end{bmatrix}}_{\mathbf{H}} \underbrace{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}}_{\mathbf{x}} + \underbrace{\begin{bmatrix} n_1 \\ n_2 \\ n_3 \end{bmatrix}}_{\mathbf{n}}$$

Thus, as can be seen from the model above, the transmit vector $\mathbf{x}$ is of dimension 2×1 , while the receive and noise vectors $\mathbf{y}$, $\mathbf{n}$ respectively are of dimension 3×1 . The above MIMO system model can also be explicitly written to describe the signal received at each receive antenna as

$$\begin{aligned} y_1 &= 2x_1 + 3x_2 + n_1 \\ y_2 &= x_1 + 3x_2 + n_2 \\ y_3 &= 4x_1 + 2x_2 + n_3 \end{aligned}$$

which basically represents a system of $r = 3$ equations for $t = 2$ unknowns x_1, x_2 . To compute the zero-forcing decoder $\mathbf{F}_{ZF}$, we first compute the matrix $(\mathbf{H}^H \mathbf{H} - 1) \mathbf{H}^H$. Observe that $\mathbf{H}^H \mathbf{H}$ can be simplified as

$$\begin{aligned}\mathbf{H}^H \mathbf{H} &= \begin{bmatrix} 2 & 1 & 4 \\ 3 & 3 & 2 \end{bmatrix} \begin{bmatrix} 2 & 3 \\ 1 & 3 \\ 4 & 2 \end{bmatrix} \\ &= \begin{bmatrix} 21 & 17 \\ 17 & 22 \end{bmatrix}\end{aligned}$$

Hence, the expression for $(\mathbf{H}^H \mathbf{H})^{-1}$ can be simplified as

$$\begin{aligned}(\mathbf{H}^H \mathbf{H})^{-1} &= \frac{1}{21 \times 22 - 17 \times 17} \begin{bmatrix} 22 & -17 \\ -17 & 21 \end{bmatrix} \\ &= \frac{1}{173} \begin{bmatrix} 22 & -17 \\ -17 & 21 \end{bmatrix}\end{aligned}$$

Thus, the zero-forcing receiver matrix $\mathbf{F}_{ZF} = (\mathbf{H}^H \mathbf{H})^{-1} \mathbf{H}^H$ can be expressed as

$$\begin{aligned}\mathbf{F}_{ZF} &= (\mathbf{H}^H \mathbf{H})^{-1} \mathbf{H}^H \\ &= \frac{1}{173} \begin{bmatrix} 22 & -17 \\ -17 & 21 \end{bmatrix} \begin{bmatrix} 2 & 1 & 4 \\ 3 & 3 & 2 \end{bmatrix} \\ &= \frac{1}{173} \begin{bmatrix} -7 & -29 & 54 \\ 29 & 46 & -26 \end{bmatrix} \\ &= \begin{bmatrix} -0.04 & -0.17 & 0.31 \\ 0.17 & 0.27 & -0.15 \end{bmatrix} \tag{6.5}\end{aligned}$$

Thus, the zero-forcing estimate of the transmit vector is given as

These expressions can now be explicitly written for the estimate of each of the transmit symbols $\hat{x}_1, \hat{x}_2$ as

$$\hat{x}_1 = -0.04y_1 - 0.17y_2 + 0.31y_3$$

$$\hat{x}_2 = +0.17y_1 + 0.27y_2 - 0.15y_3$$

It can also be confirmed that the zero-forcing receiver matrix $\mathbf{F}_{ZF}$ is of dimension 2×3 , i.e., $t \times r$. One of the chief disadvantages of the zero-forcing receivers is noise amplification. This can be understood as follows. Consider the SISO wireless system for $r = t = 1$. We have the system model given as

$$y = hx + n$$

Hence, the zero-forcing receiver is given as $f_{ZF} = (h^*h)^{-1}h^* = h^{-1} = \frac{1}{h}$. Hence, the zero-forcing estimate of the transmitted symbol x is given as

$$\begin{aligned}\hat{x}_{ZF} &= f_{ZF} \times y \\ &= \frac{1}{h} (hx + n) \\ &= x + \frac{n}{h}\end{aligned}\tag{6.6}$$

$$\begin{aligned}\hat{\mathbf{x}}_{ZF} &= \mathbf{F}_{ZF}\mathbf{y} \\ &= (\mathbf{H}^H\mathbf{H})^{-1}\mathbf{H}^H\mathbf{y} \\ &= \begin{bmatrix} -0.04 & -0.17 & 0.31 \\ 0.17 & 0.27 & -0.15 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}\end{aligned}$$

Observe that if the fading coefficient h is close to zero then the factor n/h is very high. This is termed noise amplification, which significantly distorts the performance of the receiver. The MMSE receiver which is proposed next addresses this problem.